Qiaoya Wu

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Education

University of Illinois Urbana-Champaign (UIUC),
Ph.D. in Astronomy — Aug 2021 – present

Expected graduation: May 2027

Advisor: Prof. Yue Shen

Research areas: Quasar spectral surveys; AGN accretion physics and photoionization

Xiamen University (XMU), B.S. in Astronomy — Sep 2017 – Jun 2021

Advisors: Prof. Hao-Ran Yu and Prof. Jianfeng Wu

Research areas: Cosmological N-body simulations; Black hole binaries

Thesis: Spin mode reconstruction in Lagrangian space

Fellowships and Awards

Chu Excellence in Research Graduate Student Award, UIUC	Apr 2025
Lewis E. Snyder Graduate Student Travel Award, UIUC	Mar 2025
CAPS Graduate Fellowship, Center for AstroPhysical Surveys, UIUC	2022–2023
Outstanding Undergraduate Student Award, XMU	Jun 2021
Caiwenzhong Fellowship, College of Physics Science and Technology, XMU	Apr 2020
Academic Excellence Scholarship, Department of Astronomy, XMU	Oct 2019
Guangqi Fellowship, Shanghai Astronomical Observatory, CAS	May 2019
National Scholarship, PRC Ministry of Education	Nov 2018

Publications

NASA ADS records (Oct 2026): total citations 600+ (h-index 9).

First-author and major-contribution papers

- **Wu, Q.**, Shen, Y., Done, C., Goad, M. R., & Hagen, S. (2026). *Understanding the Broad-line Region of Active Galactic Nuclei with Photoionization. II. Slim Disks, Self-shadowing, and BLR Sizes.* arXiv preprint arXiv:2606.00992 (submitted).
- **Wu, Q.**, Shen, Y., Guo, H., et al. (2025). *Understanding the Broad-line Region of Active Galactic Nuclei with Photoionization. I. The Moderate-Accretion Regime.* The Astrophysical Journal, 980(1), 134.
- **Wu, Q.**, & Shen, Y. (2023). *Improved Redshifts for DESI EDR Quasars.* Research Notes of the AAS, 7(9), 190.
- **Wu, Q.**, & Shen, Y. (2022). *A Catalog of Quasar Properties from Sloan Digital Sky Survey Data Release 16.* The Astrophysical Journal Supplement Series, 263(2), 42.

· Zheng, W. M., **Wu, Q.**, Wu, J., et al. (2022). *The Disk Veiling Effect of the Black Hole Low-mass X-Ray Binary A0620-00**. The Astrophysical Journal, 925(1), 83.

· **Wu, Q.**, Yu, H. R., Liao, S., & Du, M. (2021). *Spin Mode Reconstruction in Lagrangian Space*. Physical Review D, 103(6), 063522.

Other contributing-author papers

· Buchner, J., et al. (2026). *A large population of over-massive black hole quasars at* $z = 0.3$–0.8 *revealed by eROSITA*. arXiv preprint arXiv:2603.22425.

· SDSS Collaboration, et al. (2026). *The Nineteenth Data Release of the Sloan Digital Sky Survey*. The Astrophysical Journal Supplement Series, 285(1), 9.

· Aydar, C., et al. (2026). *Robust active galactic nucleus and host-galaxy decomposition in optical spectral fitting*. Astronomy & Astrophysics, 710, A141.

· Zhao, X., et al. (2026). *Transient Relativistic Iron Emission Line from an X-Ray Flaring Supermassive Black Hole*. The Astrophysical Journal, 1000(1), 28.

· Zhuang, M., et al. (2026). *NEXUS: A Spectroscopic Census of Broad-line AGNs and Little Red Dots at* $3 \lesssim z \lesssim 6$. The Astrophysical Journal, 999(1), 31.

· Kollmeier, J. A., et al. (2026). *Sloan Digital Sky Survey. V. Pioneering Panoptic Spectroscopy*. The Astronomical Journal, 171(1), 52.

· Zhao, X., et al. (2025). *Fast X-Ray Variability from the Coronae of Supermassive Black Holes*. The Astrophysical Journal Letters, 995(2), L72.

· Cheng, X., et al. (2025). *Quest for a Coherent Definition of Weak-line Quasars and Its Physical Implications*. The Astrophysical Journal, 994(2), 213.

· Nandra, K., et al. (2025). *The eROSITA Final Equatorial Depth Survey (eFEDS): The hard X-ray selected sample*. Astronomy & Astrophysics, 693, A212.

· Waddell, S. G. H., et al. (2024). *The eROSITA Final Equatorial Depth Survey (eFEDS): Complex absorption and soft excesses in hard X-ray-selected active galactic nuclei*. Astronomy & Astrophysics, 690, A132.

· Musiimenta, B., et al. (2023). *A new discovery space opened by eROSITA: Ionised AGN outflows from X-ray selected samples*. Astronomy & Astrophysics, 679, A84.

· Cheng, S., et al. (2020). *CUBE: Towards an Optimal Scaling of Cosmological N-body Simulations*. In 2020 20th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID) (pp. 685–690). IEEE.

· Wen, J., et al. (2019). *GRID: a student project to monitor the transient gamma-ray sky in the multi-messenger astronomy era*. Experimental Astronomy, 48, 77–95.

Approved Proposals

Wu, Q. (PI), Shen, Y. *A Local UV Benchmark for Interpreting Unobscured Coronal-Line AGNs.* 19 orbits, **HST** Cycle 34, HST-GO-18341.

Wu, Q. (PI), Shen, Y. *HST UV Spectroscopy of High-accretion-rate AGNs and the Origin of the Offset in the Broad-Line Region Size-Luminosity Relation.* 19 orbits, **HST** Cycle 31, HST-GO-17433.

Wu, Q. (PI), Wu, J., Sai, H. *Long-Term Optical Monitoring on the Black Hole Binary MAXI J1820+070.* 120 hours, **LCOGT** 2021 CTAP, CTAP2021-A0019.

Talks

Invited talks

- **WHU AGN/Galaxy Seminar**, "Linking AGN Accretion Flows and Broad-Line Region Properties through Photoionization Modeling", Wuhan University Jul 2026
- **XMU Black Hole Seminar**, "Linking AGN Accretion Flows and Broad-Line Region Properties through Photoionization Modeling", Xiamen University Jul 2026
- **UMich Galaxies Seminar**, "Understanding the AGN Line-Emitting Regions: From Optical Spectral Statistics to Broad-Line Region Photoionization Modeling", University of Michigan Feb 2026
- **UIUC Journal Club**, "Understanding the Broad-Line Region of AGN with Photoionization", UIUC Apr 2025
- **Center for AstroPhysical Surveys Seminar**, "Photoionization in the Quasar Broad-Line Region", UIUC Aug 2023
- **UIUC Journal Club**, "Quasar Spectral Decompositions from Public Surveys", UIUC May 2023
- **AGN Demographics Working Group**, "Tutorial on PyQSOFit", MPE (remote) Nov 2022

Contributed talks

- **SDSS-V 2025 Collaboration Meeting**, "From Quasar Spectra to Structure: SDSS-V Quasar Spectral Analysis and AGN Photoionization Studies", MPIA (remote) Jun 2025
- **AGN Surveys Workshop**, "Understanding the Broad-Line Region of AGN with Photoionization", Munich Institute for Astro-, Particle and BioPhysics Mar 2025
- **SDSS-V 2024 Collaboration Meeting**, "Quasar Value-Added Catalog and Visual Inspection", New Mexico State University Jun 2024
- **SDSS-V 2023 Collaboration Meeting**, "AGN Demographics", Simons Foundation, New York Aug 2023
- **The 23rd Guoshoujing Conference**, "Correlations between Halo Spins and Primordial Perturbations", Zhejiang University May 2021
- **Binary System Seminar**, "Dynamical Measurements of the Black Hole Binary MAXI J1820+070", Nanjing University Nov 2020

Outreach

Mentor, UIUC Society for Equity in Astronomy mentoring program	Fall 2022 – Spring 2024
Volunteer, UIUC girls' astronomy summer camp	Summer 2022
President, Xiamen University Astronomical Club	2018–2019
Video editor and translator, Mufu Astronomy Team	2018–2020